

Finals Essay Task 1.

Assessment on your Understanding as to what really matters to become Data Analysts.

And How to integrate SQL into your Data Analysis

Instruction:

Read the attached Article and video link before answering the questions.

Video link: <https://youtu.be/hGCKdMsfU7g>

Article Link:

<https://drive.google.com/file/d/1FhanZgD5YFiBJP0088UbEWaN7DEFbgiE/view?usp=sharing>

Following the video that you have watch and article that I ask you to read please answer the ff: question using your own words in 2–3 sentences based on the provided source material.

1. What are the four primary factors that will differentiate a data analyst candidate in 2026?

According to the video, the four main factors that will differentiate data analyst candidates in 2026 are domain knowledge, understanding the technical ecosystem, strong soft skills, and having a clear personal reason or motivation. These go beyond basic technical skills like Excel or SQL. Candidates who combine these areas can create stronger business impact and stand out in the age of AI.

2. How does domain knowledge provide a competitive advantage over purely technical candidates?

Domain knowledge helps analysts understand real business problems instead of only focusing on tools and coding tasks. It allows them to interpret data correctly and suggest solutions that support decision-making. Because of this, employers value candidates who understand the industry context, not just the technology.

3. What is the "lean tech stack" recommended for a foundational data analyst role?

The video recommends a lean tech stack that includes Excel, SQL, and one visualization tool such as Tableau or Power BI. Excel is used for pivot tables and lookup functions, SQL is used for joins and advanced queries, and visualization tools help present insights clearly. These core tools are enough to build a strong foundation in the first few months.

4. Distinguish between "learning projects" and "showing projects."

Learning projects focus on practicing skills using clean datasets to understand technical concepts. Showing projects involve working with messy, real-world data to answer actual business questions. These projects demonstrate practical thinking and are more effective when presenting your abilities to employers.

5. According to the roadmap, why is Python often deprioritized for mid-career data analyst positions?

The roadmap focuses first on mastering core tools like Excel, SQL, and visualization platforms because they are used more frequently in many analyst roles. Python is helpful but not always required early or in some mid-career analyst positions. Instead, the video emphasizes building business understanding, communication skills, and project impact before adding more advanced programming tools.

Questions on SQL-Base on the Article Posted

6. Why is establishing the "Business Context" essential for an analyst? What is a Business Context?

Business context refers to understanding the purpose behind the data and the decision that needs to be supported. It ensures that analysis answers the right question instead of producing irrelevant results. Without context, even correct queries may lead to meaningless conclusions.

7. According to the text, how does SQL function as a "flashlight"?

SQL acts like a flashlight because it helps analysts explore hidden patterns inside large datasets step by step. It allows them to investigate problems, check assumptions, and uncover useful insights. Instead of guessing, analysts use SQL to clearly see what the data contains.

8. Why is the validation step considered a professional necessity rather than an optional task?

Validation ensures that results are accurate, consistent, and trustworthy before they are shared with stakeholders. Small mistakes in queries can lead to incorrect business decisions if not checked carefully. Professionals validate their work to maintain reliability and credibility.

9. What defines "good SQL" according to the author's standards?

Good SQL is clear, efficient, and easy for others to understand and review. It produces accurate results while following logical structure and proper formatting. It also supports verification and can be reused or modified when needed.

10. Describe the fundamental mindset shift required to move from being a student to being an analyst.

Students usually focus on completing exercises correctly, but analysts focus on solving real business problems with data. The mindset shifts from writing queries for practice to asking meaningful questions and supporting decisions. Analysts think about impact, accuracy, and usefulness rather than just correctness.

